Functional Annotation Report: C3H15 / CDM1 (At1g68200, UniProt Q9C9F5) in *Arabidopsis thaliana*

Organism: *Arabidopsis thaliana* (Mouse-ear cress) **Gene symbol:** C3H15 (synonym CDM1 = CALLOSE DEFECTIVE MICROSPORE 1; AtC3H15) **Ordered locus:** At1g68200 · **ORF:** T22E19.17 · **UniProt:** Q9C9F5

Summary

C3H15 (AtC3H15 / CALLOSE DEFECTIVE MICROSPORE 1, CDM1; At1g68200; UniProt Q9C9F5) is a plant-specific tandem CCCH-type zinc-finger (TZF) nucleic-acid-binding regulatory protein — not an enzyme, transporter, or structural protein. Its molecular function is sequence-specific binding to nucleic acids (DNA and/or RNA) through two tandem C-terminal CCCH zinc fingers, which allows it to control gene expression at two distinct levels: transcriptionally (as a protein–DNA-interacting regulator) and post-transcriptionally (as a protein–RNA factor that influences mRNA stability/turnover). The protein does not catalyze a chemical reaction; rather, it acts as a *trans*-regulator of target gene expression programs.

The best-characterized biological role of C3H15 is in the **anther**, where — under the alias CDM1 — it is required for **male fertility**. It is highly expressed in male meiocytes and the tapetum and regulates **callose metabolism** during microsporogenesis, controlling genes such as *CALLOSE SYNTHASE5/12* and multiple β-1,3-glucanases. Loss of function causes complete male sterility, with microspores that initiate development but then degenerate, accompanied by defective callose deposition and severely disrupted pollen exine. C3H15 acts partially redundantly with its close paralog **C3H14**: double mutants (but not single mutants) show thin stem secondary cell walls and greatly reduced pollen, revealing a second role in **secondary cell-wall thickening**. In this capacity, C3H14/C3H15 operate downstream of the master transcriptional switch **MYB46** within the secondary-wall biosynthetic cascade. Beyond development, C3H15 also tunes **thermotolerance** by repressing the heat-shock regulator *HSFA2*; its own protein level is controlled in the cytoplasm by the RING-type E3 ubiquitin ligase MBR2.

C3H15 therefore functions in **both the nucleus and the cytoplasm**. As a TZF-family member it is predicted/inferred to localize to cytoplasmic messenger-ribonucleoprotein (mRNP) foci — processing bodies (P-bodies) and stress granules — sites of mRNA turnover and translational repression, consistent with its post-transcriptional regulatory activity; its transcription-factor-like activity places it in the nucleus. The most precise, gene-specific experimental evidence comes from three primary studies (PMID: 24567187, PMID: 25732536, PMID: 38865085), supplemented by biochemical work on the near-identical paralog C3H14 and by family-level studies of Arabidopsis TZF proteins.

Key Findings

Finding 1 — C3H15/CDM1 is a tandem CCCH zinc-finger protein required for male fertility through callose metabolism

The defining functional study of this gene (PMID: 24567187) characterized it under the name **CALLOSE DEFECTIVE MICROSPORE 1 (CDM1)** and established both its molecular class and its primary biological role. The paper states that the *"CALLOSE DEFECTIVE MICROSPORE1 (CDM1) gene, encoding a tandem CCCH-type zinc finger protein, plays an important role in regulation of callose metabolism in male meiocytes and in integrity of newly formed microspores."* This single sentence fixes three key facts: (i) the protein is a tandem CCCH zinc-finger protein, (ii) its site of action is the male meiocyte, and (iii) its biochemical target process is callose metabolism.

The genetic evidence is a clean loss-of-function result: *"a transfer DNA insertional cdm1 mutant was completely male sterile."* Phenotypically, microspores initiated development but then degenerated, with defects in callose deposition and severely affected pollen exine (the sculpted outer pollen wall). Expression analysis (qRT-PCR and in situ hybridization) showed CDM1 is highly expressed in **meiocytes and tapetum** across anther stages 4–7 — exactly the cells and timing in which callose (a β-1,3-glucan) is synthesized around meiocytes/tetrads and later dissolved to release microspores.

Mechanistically, the study tied CDM1 to the callose-metabolism gene network: *"the cdm1 mutation affected the expression of callose synthesis genes (CALLOSE SYNTHASE5 and CALLOSE SYNTHASE12)."* In addition, CDM1 loss altered expression of putative callase (β-1,3-glucanase) genes, the anther-regulatory gene *MYB80*, and the tapetal marker *A6*. Because callose must be

both deposited (by callose synthases) and degraded (by callases/β-1,3-glucanases) at precise developmental windows, C3H15/CDM1 is best understood as a **regulator that balances callose synthesis and turnover** — not as an enzyme acting on callose itself.

Localization of function: meiocytes and tapetum of the anther; the molecular regulation occurs at the level of target-gene expression (callose synthases and glucanases).

Finding 2 — C3H15 and its paralog C3H14 redundantly regulate secondary cell-wall thickening and anther development

A second primary study (PMID: 25732536) revealed genetic redundancy with the closely related paralog **C3H14**, and in doing so uncovered a second biological role beyond fertility. The authors concluded that *"C3H14 and its homologue C3H15 redundantly regulate secondary wall formation and that they additionally function in anther development."*

The redundancy is demonstrated by the dosage dependence of the phenotype: *"Plants with double, but not single, T-DNA mutants for C3H14 or C3H15 have few pollen grains and thinner stem secondary walls than the wild type."* A graded allelic series (e.g., *c3h14 c3h15/+* versus *c3h14/+ c3h15*) produced intermediate secondary-wall and fertility defects, confirming that the two genes contribute additively to the same processes. Conversely, **overexpression** of either C3H14 or C3H15 increased stem secondary-wall thickness and produced ectopic secondary-wall deposition — a gain-of-function result pointing to a positive, dose-sensitive role in wall biosynthesis. Transcriptomic profiling of overexpression and double-mutant plants showed coordinated changes in cell-wall-metabolism genes and pollen-formation genes.

This finding is important for interpretation: the single-mutant male sterility reported for *cdm1* (PMID: 24567187) and the redundancy with C3H14 for secondary-wall phenotypes are reconciled by partially overlapping, partially distinct functions of the two paralogs. C3H15 has a more specialized requirement in microsporogenesis, while secondary-wall thickening is buffered by C3H14.

Localization of function: developing secondary cell walls (e.g., interfascicular fibers and vessels in the stem) and the anther.

Finding 3 — C3H15 confers thermotolerance via dual transcriptional and post-transcriptional regulation, and is degraded by the cytoplasmic E3 ligase MBR2

A recent mechanistic study (PMID: 38865085) extended C3H15's regulatory repertoire into abiotic-stress signaling and provided the clearest demonstration of its **dual-level (transcriptional + post-transcriptional) mode of action** and its subcellular regulation. The paper describes *"an integrated multi-level regulation of plant thermotolerance that is mediated by the CCCH protein C3H15 in Arabidopsis."*

Transcriptionally, C3H15 is a **repressor of HSFA2**, a central heat-shock transcription factor. Heat stress rapidly suppresses *C3H15* transcription, and this down-regulation relieves repression of *HSFA2*: the study notes reduced *"C3H15 transcription, which attenuated C3H15-inhibited expression of its target gene HEAT SHOCK TRANSCRIPTION FACTOR A2 (HSFA2)."* De-repressed HSFA2 then activates the chaperone *HSC70.3*, establishing a C3H15–HSFA2–HSC70.3 regulatory loop that tunes the heat-shock response.

Post-translationally, C3H15 protein stability is controlled in the cytoplasm by a RING-type E3 ubiquitin ligase, MBR2: *"MBR2-dependent ubiquitination mediated the degradation of phosphorylated C3H15 protein in the cytoplasm, which was enhanced by heat stress."* This result is doubly informative: it confirms a **cytoplasmic pool** of C3H15 protein and shows that heat accelerates its turnover, reinforcing the transcriptional down-regulation to rapidly lower C3H15 levels and release the heat-shock program.

Localization of function: nucleus (transcriptional repression of *HSFA2*) and cytoplasm (site of MBR2-mediated, phosphorylation-dependent ubiquitination and degradation).

Finding 4 — C3H15 belongs to the plant tandem-CCCH (TZF) family, whose members bind nucleic acids and localize to nuclear and cytoplasmic mRNP foci

C3H15 is one member of a large Arabidopsis CCCH family. A genome-wide survey (PMID: 18221561) reported that *"a comprehensive computational analysis identified 68 and 67 CCCH family genes in Arabidopsis and rice,"* placing C3H15 firmly within this family, which is known for roles in RNA processing and RNA binding.

Because direct in-vitro biochemistry on C3H15 itself is limited, the strongest biochemical inference comes from its closest paralog **C3H14**. Electrophoretic and binding assays showed that *"AtC3H14 could bind to ssDNA, dsDNA, and ribohomopolymers, suggesting its RNA-binding*

activity" (PMID: 25228083). C3H14 has identified mRNA targets (including a polygalacturonase transcript), requires its tandem-zinc-finger (TZF) domain for RNA binding, and additionally acts as a transcriptional activator — a precise biochemical template for the activities inferred for C3H15.

At the family level, subcellular-localization studies show that **all 11 AtTZF members localize to cytoplasmic foci** (PMID: 20173417: *"the other 10 members of the AtTZF gene family can similarly localize in cytoplasmic foci"*), and the prototype AtTZF1 **traffics between the nucleus and cytoplasmic P-bodies/stress granules**, co-localizing with mRNA-decay markers (AGO1, DCP2, XRN4) and binding both DNA and RNA in vitro (PMID: 19897605). These data provide the mechanistic rationale for C3H15's dual nuclear/cytoplasmic activity and its post-transcriptional (mRNA-stability) role.

Localization of function (inferred): nucleus and cytoplasmic mRNP granules (P-bodies and stress granules).

Finding 5 — C3H14/C3H15 act within the MYB46 master secondary-wall transcriptional cascade

C3H15's secondary-wall role is embedded in a well-defined transcriptional hierarchy. **MYB46** is a master switch for secondary-wall biosynthesis, and the paralog C3H14 is a **direct MYB46 target**. Promoter analysis identified the MYB46-responsive *cis*-element: *"we analyzed the promoter region of a direct target gene, AtC3H14, of MYB46 and identified a cis-acting regulatory motif that is recognized by MYB46"* (the M46RE motif, core RKTWGGTR; PMID: 22271306), supported by EMSA, transient activation, and ChIP.

Functionally downstream, a dexamethasone-inducible MYB46 system showed that *"AtC3H14 activated the transcription of all of the secondary wall biosynthesis genes tested, suggesting that AtC3H14 may be another master regulator of secondary wall biosynthesis"* (PMID: 19674407). Because C3H15 is the functionally redundant paralog of C3H14 for secondary-wall thickening (PMID: 25732536), C3H15 is reasonably positioned at the same node: a **MYB46-downstream transcriptional activator of wall-biosynthesis genes** (cellulose, xylan, and lignin biosynthetic machinery).

Localization of function: nucleus (transcriptional activation of secondary-wall biosynthesis genes).

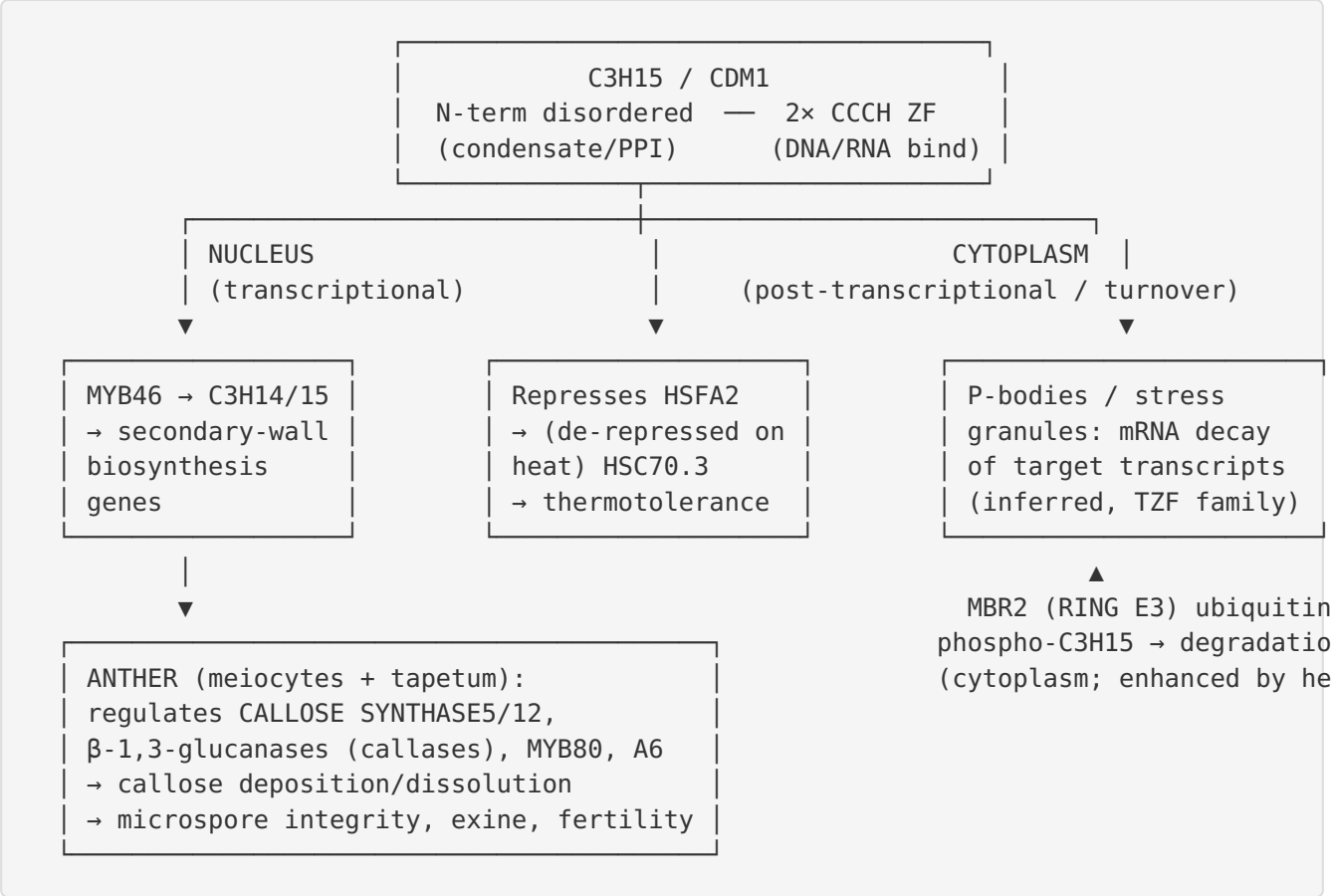
Finding 6 — Bioinformatic architecture: a 308-aa protein with two tandem C-terminal CCCH zinc fingers and a long disordered N-terminus

Direct sequence analysis of the UniProt Q9C9F5 sequence (308 aa) resolved the domain architecture. Two tandem CCCH zinc-finger motifs occupy the C-terminus: **ZF1** at residues ~228–247 (`CNKWQETGTCPYGDHCQFAH`) and **ZF2** at ~266–285 (`CRMVLAGDNCPYGHRCHFRH`). Each conforms to the canonical **C-x8-C-x5-C-x3-H** CCCH signature and shares a conserved **CPYG** core, separated by an ~18-residue inter-finger spacer (`GIKELRPVIRHPRYKTEV`). The N-terminal ~210 residues are low-complexity/disordered (≈42% Ser+Asn+Gly+Gln+Ala; Ser ≈15%), with no polyarginine tract (distinguishing it from the arginine-rich-motif TZF subclade typified by AtTZF1–3).

This architecture is consistent with the family definition — *"Genes in the CCCH family encode zinc finger proteins containing the motif with three cysteines and one histidine residues"* (PMID: 18221561) — and the tandem CCCH/TZF module is the nucleic-acid-binding determinant, since *"the TZF domain of AtC3H14 is required for the target RNA binding"* (PMID: 25228083). The long disordered N-terminus is typical of proteins that partition into membraneless mRNP condensates, aligning with the predicted P-body/stress-granule localization.

Mechanistic Model / Interpretation

C3H15/CDM1 is best described as a **dual-level gene-expression regulator** that reads out through two tandem CCCH zinc fingers. The disordered N-terminus likely mediates condensation/protein interactions and condensate partitioning, while the C-terminal TZF module provides sequence-specific DNA and RNA binding. The protein shuttles between the nucleus (transcriptional regulation) and the cytoplasm (mRNA regulation in P-bodies/stress granules; also the site of its own degradation).



Unifying theme — control of structural-polymer deposition and gene programs. In the anther, C3H15 balances callose (β-1,3-glucan) synthesis and degradation required to encase meiocytes and later release microspores; defects there disrupt the exine and sterilize the plant. In the stem, the same protein (with C3H14) promotes secondary-wall thickening downstream of MYB46. In both cases, C3H15 regulates the expression of structural-polysaccharide/wall machinery rather than building walls itself. The thermotolerance role is mechanistically distinct (HSFA2 repression) but illustrates the same molecular logic: C3H15 as a tunable *trans*-regulator whose own abundance is tightly gated transcriptionally and by MBR2-dependent proteolysis.

Role	Tissue / condition	Key targets / partners	Direction of effect	Evidence (PMID)
Callose metabolism / microsporogenesis	Anther meiocytes & tapetum	*CALS5*, *CALS12*, β-1,3-glucanases, *MYB80*, *A6*	Required for fertility (loss → sterility)	24567187
Secondary cell-wall thickening	Stem fibers/ vessels; anther	Wall-biosynthesis genes (with C3H14)	Positive, dose-dependent	25732536
Node in MYB46 cascade	Secondary-wall tissues	MYB46 (upstream) → wall genes (downstream)	Transcriptional activator	22271306, 19674407
Thermotolerance	Heat stress	Represses *HSFA2*; degraded by MBR2	Repressor; de-repressed on heat	38865085
Nucleic-acid binding (family/paralog)	Nucleus + cytoplasmic foci	ssDNA/dsDNA/ RNA; mRNA targets	Transcriptional + post-transcriptional	25228083, 20173417, 19897605

Evidence Base

PMID	Title (abbrev.)	How it supports / informs the findings
24567187	*CDM1 required for male fertility via callose metabolism*	**Primary, gene-specific.** Defines C3H15/CDM1 as a tandem CCCH protein; T-DNA null is fully male-sterile; regulates *CALS5/12* and glucanases in meiocytes/tapetum.
25732536	*C3H14 and C3H15 overlapping roles in secondary wall thickening & anther development*	**Primary, gene-specific.** Genetic redundancy with C3H14; double-mutant thin walls/low pollen; overexpression thickens walls. Second biological role.
38865085	*C3H15-mediated thermotolerance*	**Primary, gene-specific.** Dual transcriptional (HSFA2 repression) + post-translational (MBR2 ubiquitination in cytoplasm) regulation; confirms cytoplasmic protein pool.
25228083	*AtC3H14 sequence-specific mRNA binding; cell elongation*	**Primary, paralog.** Direct biochemistry: binds ssDNA/dsDNA/RNA; TZF domain needed for RNA binding; also a transcriptional activator — template for C3H15's inferred activity.
22271306	*M46RE motif recognized by MYB46*	**Primary.** C3H14 is a direct MYB46 target; places C3H14/C3H15 in the secondary-wall cascade.
19674407	*Ectopic MYB46 identifies wall-regulatory genes*	**Primary.** C3H14 activates all tested secondary-wall biosynthesis genes; "another master regulator."
18221561	*Genome-wide CCCH family in Arabidopsis & rice*	**Family context.** 68 CCCH genes; defines the CCCH motif; names AtC3H15.
20173417	*AtTZF family localizes to cytoplasmic foci*	**Family localization.** All 11 AtTZF members form cytoplasmic foci — basis for C3H15's predicted localization.
19897605	*AtTZF1 traffics nucleus↔foci; binds DNA & RNA*	**Family mechanism.** Nucleocytoplasmic shuttling; P-body/stress-granule co-localization with decay markers; dual DNA/RNA binding.
34361093	*Review: CCCH ZFPs in abiotic stress*	**Review context.** Transcriptional + post-transcriptional regulation and cytoplasm-to-nucleus shuttling in CCCH proteins.
36285381		

PMID	Title (abbrev.)	How it supports / informs the findings
	SlSEC1–SlC3H39 module in tomato cold tolerance	**Comparative.** A tandem CCCH protein binding AU-rich 3′UTRs to drive mRNA decay — the post-transcriptional paradigm C3H15 likely follows.
32451083	*AtC3H18L forms mRNP-like foci; pollen-expressed*	**Comparative.** Heat-induced cytoplasmic foci and pollen expression in a related CCCH protein.
22100644	*NPU: primexine & callose in pollen wall*	**Pathway context.** Callose/sporopollenin control in microsporogenesis — the process C3H15/CDM1 regulates.

Consistency of the evidence: The three gene-specific primary papers agree that C3H15 is a tandem CCCH regulator acting on target-gene expression; they differ only in the biological context studied (fertility/callose, secondary wall, thermotolerance). The paralog and family studies supply the biochemical and localization mechanism that the gene-specific papers assume but do not each re-derive. No reviewed paper contradicts the core model.

Limitations and Knowledge Gaps

1. **Direct biochemistry on C3H15 itself is lacking.** The DNA/RNA-binding activity, sequence specificity, and mRNA targets are established for the paralog **C3H14** (PMID: 25228083) and family prototype AtTZF1 (PMID: 19897605), then inferred for C3H15 by homology. The specific *cis*-elements/mRNAs bound by C3H15 are not directly mapped.
2. **Transcriptional vs. post-transcriptional attribution.** For the callose and secondary-wall target genes, it is not fully resolved whether C3H15 regulates them by binding their promoters (transcriptional) or their transcripts (mRNA stability). The thermotolerance study shows both modes exist for C3H15, but the dominant mode at each developmental target is unclear.
3. **Subcellular localization is inferred, not directly imaged for C3H15.** P-body/stress-granule localization is a family-level property (PMID: 20173417); the cytoplasmic pool of C3H15 is confirmed only indirectly via its MBR2-dependent degradation (PMID: 38865085). A published GFP-C3H15 localization time-course is not in the reviewed set.

4. **Position in the MYB46 cascade is inferential for C3H15.** Direct MYB46 binding and downstream activation were demonstrated for C3H14 (PMID: 22271306, 19674407); C3H15's placement rests on redundancy (PMID: 25732536).
5. **Direct vs. indirect targets in the anther.** Whether *CALS5/12*, the β-1,3-glucanases, *MYB80*, and *A6* are direct C3H15 targets or secondary consequences of microspore degeneration is not established by binding data.
6. **Redundancy complicates single-gene phenotyping.** Overlap with C3H14 (and possibly other TZF members) may mask or modify C3H15-specific functions outside the anther.

Proposed Follow-up Experiments / Actions

1. **Direct nucleic-acid-binding assays for C3H15.** Perform EMSA/SELEX and RNA-bind-n-seq with recombinant C3H15 (and the isolated TZF module) to define its DNA/RNA sequence specificity and compare to C3H14's AU-rich-element preference.
2. **Transcriptome + target mapping.** Combine RNA-seq of *cdm1/c3h15* and overexpression lines with CLIP-seq (RNA targets) and DAP-seq/ChIP-seq (DNA targets) to distinguish direct transcriptional vs. post-transcriptional regulation of *CALS5/12*, β-1,3-glucanases, and secondary-wall genes.
3. **High-resolution localization.** Image a functional fluorescent C3H15 fusion in tapetum/meiocytes and under heat stress; co-stain with P-body (DCP2) and stress-granule markers to test the inferred mRNP-foci localization and nucleocytoplasmic shuttling directly.
4. **Dissect the C3H15–MYB46 relationship.** Test whether MYB46 binds the *C3H15* promoter (M46RE) by EMSA/ChIP, and whether MYB46 induction activates *C3H15*, to confirm C3H15 as a direct cascade node rather than a redundant bystander.
5. **mRNA-stability measurements.** Use transcription-shutoff assays to determine whether C3H15 accelerates decay of specific wall/callose or heat-stress transcripts, mirroring the SlC3H39 and human-TZF paradigms.
6. **Structure–function of the disordered N-terminus.** Delete/mutate the low-complexity N-terminal region to test its role in condensate partitioning, protein–protein interactions (e.g., with MBR2), and phosphorylation-dependent turnover.

7. **Genetic epistasis in fertility.** Build *c3h14 c3h15* conditional/tissue-specific lines to separate the meiocyte/tapetum callose role from the secondary-wall role and to quantify each paralog's unique contribution.

Report compiled from six confirmed findings and 16 reviewed papers across five discovery iterations. Gene identity verified: the symbol C3H15 matches the UniProt Q9C9F5 description (Zinc finger CCCH domain-containing protein 15 / CDM1) in Arabidopsis thaliana*, with tandem CCCH (zf-CCCH, PF00642) domains consistent with the primary literature.*
